Coordinate Geometry

CHAPTER 3

Pair of Straight Lines **3.1–3.4**

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Ananth Garg

CHAPTER 3

Pair of Straight Lines

DPP 3.1
Pair of Straight Lines

Single Correct Answer Type

1. The value of λ such that $2x^2 - 10xy + 12y^2 + 5x + \lambda y - 3 = 0$ represents a pair of straight lines, is
   (a) -10 (b) -9
   (c) 10 (d) 9

2. If the equation $2x^2 + 2hxy + 6y^2 - 4x + 5y - 6 = 0$ represents a pair of straight lines, then the length of intercept on the x-axis cut by the lines is equal to
   (a) 2 units (b) 4 units
   (c) $\sqrt{7}$ units (d) 0 unit

3. If pair of straight lines $3x^2 - 4xy + y^2 + 8x - 2y - 3 = 0$ and line $2x - 3y + \lambda = 0$ are concurrent, then the value of λ is
   (a) $4\pi - 11$ (b) -11
   (c) $\frac{1}{11}$ (d) 11

4. A circle rolls between pair of lines $9x^2 + 24xy + 16y^2 - 25 = 0$ touching both. Then its area is
   (a) 4π sq. units (b) 8π sq. units
   (c) 12π sq. units (d) π sq. units

Multiple Correct Answers Type

5. Let $0 < p < q$ and $a \neq 0$ such that the equation $px^2 + 4\lambda xy + qy^2 + 4a(x + y + 1) = 0$ represents a pair of straight lines. Then a can lie in the interval(s)
   (a) $(-\infty, \infty)$ (b) $(-\infty, p]$
   (c) $[p, q]$ (d) $[q, \infty)$

6. If the equation $9x^2 + 2hxy + 4y^2 + 6x + 2fy - 3 = 0$ represents two parallel lines then
   (a) $h = 6, f = 2$ (b) $h = -6, f = 2$
   (c) $h = 6, f = -2$ (d) $h = -6, f = -2$

7. For the pair of lines $2x^2 + 5xy + 2y^2 + 4x + 5y + a = 0$ and the line $L : bx + y + 5 = 0$,
   (a) $a = 2$
   (b) $a = -2$
   (c) there exists no circle which touches the pair of lines and the line L if $b = 5$.
   (d) there exists no circle which touches the pair of lines and the line L if $b = -5$.

8. The equation of the diagonal of the square formed by the pairs of lines $xy + 4x - 3y - 12 = 0$ and $xy - 3x + 4y - 12 = 0$ is
   (a) $x - y = 0$ (b) $x + y + 1 = 0$
   (c) $x + y = 0$ (d) $x - y + 1 = 0$

9. If the lines in the pair $6x^2 - \alpha xy - 3y^2 - 24x + 3y + \beta = 0$ intersect on x-axis, then
   (a) $\beta = 24$ (b) $\alpha = 12$
   (c) $\alpha = 3/2$ (d) $\beta = 2/3$

10. If the equation $x^2 + y^2 + 2xy + 2gx + 2fy + 4 = 0$ represents a pair of straight lines, then the possible value(s) of g can be
    (a) -3 (b) -2
    (c) 1 (d) 5

Numerical Value Type

11. Let $\alpha x - \beta y - 8 = 0$ be the equation of a line which is parallel to the line common to the pair of lines given by $x^2 - y^2 - x + y = 0$ and $2x^2 + y^2 + 3xy - 2x - y = 0$. Also, it makes a triangle of area 8 sq. unit with the coordinate axes as well as its x and y intercepts are negative. The value of $\alpha^2 + \beta^2$ is _____.

12. Let the line $y = x$ be median of the triangle OAB where O is the origin and the equation $ax^2 + 2hxy + by^2 = 0$, $a, h, b \in N$, represents the combined equation of lines OA and OB. A and B lie on the ordinate $x = 3$. If the slope of OA is twice the slope of OB, then the greatest possible value of $a - 2h + b$ is ____.

Answer Key

Single Correct Answer Type

1. (b) **2.** (b) **3.** (d) **4.** (d)

Multiple Correct Answers Type

5. (b, d) **6.** (a, d) **7.** (a, c) **8.** (a, b) **9.** (a, c) **10.** (a, b, d)

Numerical Value Type

11. (8) **12.** (35)

Ananth Garg

DPP 3.2

Homogeneous Equation of Second Degree

Single Correct Answer Type

1. The joint equation of pair of lines which pass through the origin and are perpendicular to the lines represented by the equation $y^2 + 3xy - 6x + 5y - 14 = 0$ is

(a) $y^2 - 3xy = 0$ (b) $3y^2 - xy = 0$
(c) $x^2 - 3xy = 0$ (d) $3x^2 - xy = 0$

2. If $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ is combined equation of three straight lines and two of them are perpendicular, then sum of all the possible values of a is equal to _____.

(a) 3/8 (b) –3/4
(c) 1/4 (d) –2

3. The line $y = mx$ will bisect the angle between the lines $ax^2 + 2hxy + by^2 = 0$ if

(a) $h(1 + m^2) = m(a + b)$ (b) $h(1 - m^2) = m(a - b)$
(c) $h(1 + m^2) = m(a - b)$ (d) None of these

4. The line $x + y = 7$ forms a triangle with the lines $ax^2 + 2hxy + ay^2 = 0$, $(a \neq 0)$. The triangle is

(a) isosceles (b) scalene
(c) equilateral (d) right angled triangle

5. If the line passing through $P(1, 2)$ making an angle of 45° with the x-axis in the positive direction meets the pair of lines $x^2 + 4xy + y^2 = 0$ at A and B, then $PA \times PB =$ _____.

(a) 13/3 (b) 13/6
(c) 11/6 (d) 11/3

6. Which of the following pairs of straight lines intersect at right angle?

(a) $2x^2 = y(x + 2y)$ (b) $(x + y)^2 = x(y - 3x)$
(c) $2y(x + y) = xy$ (d) $y = \pm 2x$

Multiple Correct Answers Type

7. If $x^2 + 2hxy + y^2 = 0$ $(h \neq 1)$ represents the equations of the straight lines passing through the origin which make an angle α with the straight line $y + x = 0$, then

(a) $\sec 2\alpha = h$
(b) $\cos\alpha = \sqrt{\dfrac{1+h}{2h}}$
(c) Sum of slopes of lines $= -2 \sec 2\alpha$
(d) $\cot\alpha = \sqrt{\dfrac{h+1}{h-1}}$

8. The equation $x^2 + k_1y^2 + 2k_2y = a^2$ represents a pair of perpendicular straight lines if

(a) $k_1 = 1, k_2 = a$ (b) $k_1 = 1, k_2 = -a$
(c) $k_1 = -1, k_2 = -a$ (d) $k_1 = -1, k_2 = a$

9. A line passes through the point (2, 0). Then which of the following is/are the slope(s) of the line, for which its intercept between $y = x - 1$ and $y = -x + 1$ subtends a right angle at the origin?

(a) $-\dfrac{1}{\sqrt{3}}$ (b) $-\sqrt{3}$
(c) $\dfrac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

Answer Key

Single Correct Answer Type

1. (c) **2.** (b) **3.** (b) **4.** (a) **5.** (a) **6.** (a)

Multiple Correct Answers Type

7. (a, b, c, d) **8.** (c, d) **9.** (a, c)

Ananth Garg

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CHAPTER 3

DPP 3.1

Single Correct Answer Type

1.(b)

2. (b) Putting $y = 0$, we get

$$x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0 \Rightarrow x = -1, 3$$

So, length of intercept on x-axis $= |3 - (-1)| = 4$ units

3. (d) We have $f(x, y) \equiv 3x^2 - 4xy + y^2 + 8x - 2y - 3 = 0$

Differentiating w.r.t. x, keeping y as constant, we get

$$\frac{\partial f}{\partial x} = 6x - 4y + 8 = 0 \quad \ldots(1)$$

Differentiating w.r.t. y, keeping x as constant

$$\frac{\partial f}{\partial y} = -4x + 2y - 2 = 0 \quad \ldots(ii)$$

Solving (1) and (2), we get the point of intersection of the pair of lines as (2, 5).

If it satisfies the equation $2x - 3y + \lambda = 0$, then $\lambda = 11$.

4. (d) We have $9x^2 + 24xy + 16y^2 - 25 = 0$

$\Rightarrow (3x + 4y)^2 - 25 = 0$

$\Rightarrow 3x + 4y + 5 = 0$ or $3x + 4y - 5 = 0$

These lines are parallel and tangents to the circle.

$\therefore$ Diameter of the circle = Distance between parallel tangents

$$= \frac{|5-(-5)|}{\sqrt{3^2+4^2}} = 2 \text{ units}$$

$\therefore$ Radius of the circle = 1 unit

So, area of the circle = π sq. units

Multiple Correct Answers Type

5. (b, d)

We have $px^2 + 4\lambda xy + qy^2 + 4ax + 4ay + 4a = 0$

Comparing with $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$, we get

$a = p, h = 2\lambda, b = q, g = f = 2a, c = 4a$

If an equation represents a pair of straight lines, then we must have $abc + 2hgf - af^2 - bg^2 - ch^2 = 0$

or $p.q.4a + 2.2a.2a.2\lambda - p.4a^2 - q.4a^2 - 4a.4\lambda^2 = 0$

$\Rightarrow 4\lambda^2 - 4a\lambda + \{(p+q)a - pq\} = 0 \quad (\because a \neq 0)$

For $\lambda \in R$,

Discriminant ≥ 0

$\Rightarrow 16a^2 - 16\{(p+q)a - pq\} \geq 0 \Rightarrow (a-p)(a-q) \geq 0$

$\therefore a \leq p$ or $a \geq q$

6. (a, d)

7. (a, c)

$2x^2 + 5xy + 2y^2 + 4x + 5y + a = 0$ represents the pair of lines if

$$\begin{vmatrix} 2 & 5/2 & 2 \\ 5/2 & 2 & 5/2 \\ 2 & 5/2 & a \end{vmatrix} = 0 \Rightarrow a = 2$$

So, the equation of pair of lines is

$$2x^2 + 5xy + 2y^2 + 4x + 5y + 2 = 0.$$

Therefore, the lines are $x + 2y + 1 = 0$ and $2x + y + 2 = 0$.

These lines are concurrent with $bx + y + 5 = 0$.

$$\therefore \begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ b & 1 & 5 \end{vmatrix} = 0 \Rightarrow b = 5$$

When lines are concurrent, no circle can be drawn touching all three lines.

8. (a, b)

$xy + 4x - 3y - 12 = 0$

$\Rightarrow (x-3)(y+4) = 0 \Rightarrow x = 3$ and $y = -4$

$xy - 3x + 4y - 12 = 0$

$\Rightarrow (x+4)(y-3) = 0 \Rightarrow x = -4$ and $y = 3$

So, vertices of square are $A(-4, -4)$, $B(-4, 3)$, $C(3, 3)$ and $D(3, -4)$.

So, diagonal AC is $x = y$ and diagonal BD is $x + y + 1 = 0$.

9. (a, c)

Lines intersect on x-axis.

So, putting $y = 0$, we get

$6x^2 - 24x + \beta = 0$

This equation must have equal roots.

$\therefore (-24)^2 - 4(6)\beta = 0 \Rightarrow \beta = 24$

Differentiating w.r.t. x keeping y as constant, we get

$12x - \alpha y - 24 = 0$

Differentiating w.r.t. y keeping x as constant, we get

$-\alpha x - 6y + 3 = 0$

For x-axis $y = 0$, $x = 2$ and $x = \frac{3}{\alpha}$.

$\therefore 2 = \frac{3}{\alpha} \Rightarrow \alpha = \frac{3}{2}$

10. (a, b, d)

Given equation is $x^2 + 2x(y+g) + y^2 + 2fy + 4 = 0$

$$\therefore x = \frac{-2(y+g) \pm \sqrt{4(y+g)^2 - 4(y^2 + 2fy + 4)}}{2}$$

or $x = -(y+g) \pm \sqrt{(g^2 - 4) + 2y(g - f)}$

So, the equation will represent the pair of straight lines if $g = f$ and $g^2 - 4 \geq 0$.

$\therefore g \geq 2$ or $g \leq -2$.

Numerical Value Type

11. (8) $x^2 - y^2 - x + y = 0$

$\Rightarrow x - y = 0, x + y - 1 = 0$

$2x^2 + y^2 + 3xy - 2x - y = 0$

$\Rightarrow 2x + y = 0, x + y - 1 = 0$

So, the common line is $x + y - 1 = 0$.

The line parallel to $x + y - 1 = 0$ is $x + y = \lambda$.

This line makes a triangle of area 8 sq. units with the coordinate axes.

$\therefore \quad \frac{1}{2}\lambda^2 = 8 \quad \Rightarrow \quad \lambda = \pm 4$

So, the required line is

$x + y + 4 = 0$ (as x and y intercepts are negative)

$\Rightarrow \quad -2x - 2y - 8 = 0 \quad \Rightarrow \quad \alpha = -2, \beta = 2 \quad \Rightarrow \quad \alpha^2 + \beta^2 = 8$

12.(35) Point (3, 3) is the mid-point of AB.

Let $A \equiv (3, 3 + \alpha)$ and $B \equiv (3, 3 - \alpha)$.

$\Rightarrow \quad \frac{3+\alpha}{3} = 2\frac{3-\alpha}{3}$ (given)

$\Rightarrow \quad \alpha = 1$

$\Rightarrow \quad A \equiv (3, 4)$ and $B \equiv (3, 2)$

So, the equation of OA is $3x - 4y = 0$ and the equation of OB is $3x - 2y = 0$.

So, combined equation of OA and OB is

$(2x - 3y)(4x - 3y) = 0$

or $\quad 8x^2 - 18xy + 9y^2 = 0$

$\therefore \quad a - 2h + b = 8 + 18 + 9 = 35$

DPP 3.2

Single Correct Answer Type

1. (c) Homogeneous part of the given equation is $y^2 + 3xy = 0$, which represents the straight lines $y = 0$ and $y + 3x = 0$.

Now, lines perpendicular to these lines are $x = 0$ and $x - 3y = 0$.

So, combined equation of above lines is $x^2 - 3xy = 0$.

2. (b)

3. (b) The equation of pair of bisectors of angles between the lines $ax^2 + 2hxy + by^2 = 0$ is

$$\frac{x^2 - y^2}{xy} = \frac{a-b}{h}$$

or $\quad h(x^2 - y^2) = (a - b)xy \quad \ldots(1)$

But $y = mx$ is one of these lines.

Putting $y = mx$ into (1), we get

$h(x^2 - m^2x^2) = (a - b)x.\, mx$

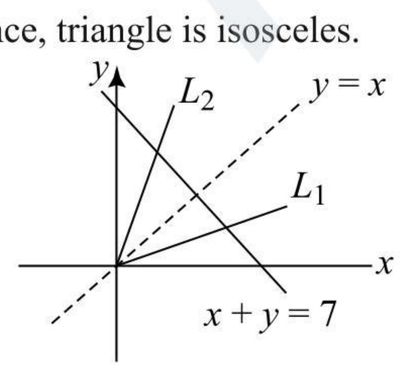
Dividing by x^2, we get

$h(1 - m^2) = m(a - b)$.

4. (a) In the equation $ax^2 + 2hxy + ay^2 = 0$, interchanging x and y, we get the same equation.

So, the component lines of the above equation are symmetrical about the line $y = x$.

Hence, triangle is isosceles.



5. (a) Coordinates of any point at distance 'r' from the point $P(1, 2)$ on the line making an angle of 45° with the x-axis are

$$\left(1 + \frac{r}{\sqrt{2}}, 2 + \frac{r}{\sqrt{2}}\right).$$

Putting these coordinates in the given equation of pair of lines, we get $3r^2 + 9\sqrt{2}\, r + 13 = 0$

Product of roots, $r_1r_2 = 13/3 = PA \times PB$

6. (a)

Multiple Correct Answers Type

7.(a, b, c, d)

Let lines $x^2 + 2hxy + y^2 = 0$ be given by $y = m_1x$ and $y = m_2x$.

$\therefore \quad m_1 + m_2 = -2h$

Slope of $y + x = 0$ is -1.

$\therefore \quad \tan\alpha = \left|\frac{m_1+1}{1-m_1}\right|$ and $\tan\alpha = \left|\frac{m_2+1}{1-m_2}\right|$

$\therefore \quad \tan\alpha = \frac{m_1+1}{1-m_1}$ and $\tan\alpha = -\left(\frac{m_2+1}{1-m_2}\right)$

(for +ve signs, both gives the same value but $m_1 \neq m_2$).

$\Rightarrow \quad m_1 = \frac{\tan\alpha - 1}{\tan\alpha + 1}, m_2 = \frac{\tan\alpha + 1}{\tan\alpha - 1}$

So, $\quad m_1 + m_2 = -2\sec 2\alpha$

$\Rightarrow \quad h = \sec 2\alpha$

$\Rightarrow \quad \cos 2\alpha = \frac{1}{h} \quad \Rightarrow \quad 2\cos^2\alpha - 1 = \frac{1}{h}$

$\Rightarrow \quad \cos\alpha = \left(\frac{1+h}{2h}\right)^{\frac{1}{2}} \Rightarrow \quad \cot\alpha = \sqrt{\frac{h+1}{h-1}}$

8. (c, d)

Lines are perpendicular if $1 + k_1 = 0$.

Also, for pair of straight lines,

$abc + 2hgf - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow \quad k_1 = -1$ and $k_2 = \pm a$

9. (a, c)

The combined equation of straight lines $y = x - 1$ and $y = -x + 1$ is

$(x - y - 1)(x + y - 1) = 0$

or $\quad x^2 - y^2 - 2x + 1 = 0 \quad \ldots(1)$

Let the equation of the line passing through (2, 0) is

$y = m(x - 2) \quad \ldots(2)$

On homogenizing equation (1) using line (2), we get

$$x^2 - y^2 - 2x\left(\frac{mx - y}{2m}\right) + \left(\frac{mx - y}{2m}\right)^2 = 0$$

This is the equation of pair of straight lines passing through the origin.

Since these lines are perpendicular, coefficient of x^2 + coefficient of $y^2 = 0$

$\Rightarrow \quad m = \pm\frac{1}{\sqrt{3}}$